

CML 4 - Linear Regression assumptions

2h offline. After fitting OLS (CML 3), ask: **when are the coefficients and intervals trustworthy?** Diagnose with residuals; fix with practical remedies.

0. Hook (5 min)

Show a residual-vs-fitted plot with a clear curve (or fan). Ask: “Would you trust this model’s p-values / CIs?”

Line to land: “Predictions can look fine while **inference and coef meaning** break. Assumptions are about the **errors** - we check them via **residuals**.”

1. The classical checklist (LINE + multicollinearity)

Visual: visuals/01_line_checklist.svg

Letter	Assumption	In plain English
L	Linearity	$E[y$
I	Independence	Errors don’t correlate across rows (no time/cluster leakage)
N	Normality	Residuals $\sim$ normal (mainly for CI / tests; less critical for large n point estimates)
E	Equal variance	Homoedasticity - residual spread doesn’t grow/shrink with fitted values
+	No perfect multicollinearity	Features aren’t perfect linear copies of each other

Model form reminder

```
y = theta_0 + theta_1 x1 + ... + theta_n xn + epsilon
```

Assumptions live mostly on **epsilon**. We never see **epsilon** - we plot **residuals** $e = y - \hat{y}$.

2. Residuals are the diagnostic language

Visual: visuals/02_residual_anatomy.svg

```
residual_i = y_i - y_hat_i
```

Core plots (board + notebook):

1. **Residuals vs fitted** - linearity + equal variance
2. **Residuals vs each key feature** - missed nonlinearity in that feature
3. **Histogram / Q-Q of residuals** - normality
4. **(Optional)** residuals vs row order / time - independence

Healthy residual-vs-fitted: random cloud around 0, roughly constant vertical spread.

3. Linearity - what breaks, how it looks, what to do

Visual: `visuals/03_linearity.svg`

Symptom	Plot cue	Remedies
Curved mean residual	Smooth U / in residual-vs-fitted or vs x	Add transforms (<code>log</code> , <code>sqrt</code>), polynomial / spline terms, or a nonlinear model

Interactive: “If residuals bend vs `median_income`, is income’s effect linear?”

4. Equal variance (homoscedasticity)

Visual: `visuals/04_homo_hetero.svg`

Symptom	Plot cue	Remedies
Fan / mega-phone	Spread grows with <code>y_hat</code>	Log-transform y (prices!), weighted least squares, robust SE (later), or a different model family

Fintech note: house prices / balances often show fans - log target is a first-line habit.

5. Normality of residuals

Visual: `visuals/05_qq_normal.svg`

- Heavy tails / skew -> CI and tests can be off
- Large n: CLT helps **means of coefs**; still report diagnostics
- Extreme outliers: investigate; don’t blindly delete

Remedies: transform y, robust regression, or accept prediction-only use of OLS.

6. Independence (short but sharp)

Red flag	Example	Fix
Time series / panels	Same user over months	Time-aware split; cluster SE; lagged features
Grouped data	Multiple rows per household	Cluster at that level

CML 2 echo: leakage and bad splits create **fake** independence.

7. Multicollinearity

Visual: visuals/06_vif.svg

- Perfect collinearity -> $X^T X$ singular (closed form fails)
- Near-collinearity -> coefs wild, huge SE, signs flip when you add a cousin feature

Checks

```
# Correlation heatmap among features  
# VIF (rule of thumb): VIF > 5-10 -> investigate
```

```
VIF_j = 1 / (1 - R^2_j)
```

where R^2_j is from regressing feature j on the other features.

Remedies: drop / combine features, PCA (later), regularization (Session 7).

8. Live lab - diagnose the house-price OLS

Notebook: [CML4_LR_Assumptions.ipynb] (CML4_LR_Assumptions.ipynb)

Reuse CML 3 Pipeline -> then:

```
fit -> residuals = y_test - y_pred  
-> residual vs fitted  
-> residual vs median_income (scaled)  
-> Q-Q / hist  
-> feature correlation / VIF sketch
```

Ask the room: which assumption looks most violated on California housing?